# Financing Activity Intensity and the Cross Section of Stock Returns

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**Abstract**

This paper studies the asset pricing implications of Financing Activity Intensity (FAI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on FAI achieves an annualized gross (net) Sharpe ratio of 0.54 (0.49), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 24 (27) bps/month with a t-statistic of 2.10 (2.38), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Asset growth, Growth in book equity, change in net operating assets, Inventory Growth, Change in equity to assets, Change in financial liabilities) is 24 bps/month with a t-statistic of 2.22.

# 1 Introduction

Financial markets process vast amounts of information daily, yet the speed and completeness of this information incorporation remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, mounting evidence demonstrates that markets exhibit predictable patterns arising from systematic frictions in the information diffusion process. These frictions manifest particularly strongly when investors face constraints on attention and information processing capacity, leading to delayed price adjustments and return predictability.

Despite extensive research documenting various forms of market inefficiency, our understanding of how corporate financing activities influence stock returns through information channels remains incomplete. Financing decisions represent fundamental corporate events that signal management's private information about firm prospects, yet the complexity and infrequency of these activities may cause investors to underreact to their implications. This paper examines whether Financing Activity Intensity (FAI) predicts cross-sectional stock returns, focusing specifically on how limited investor attention and gradual information diffusion create systematic mispricing that generates predictable return patterns.

The slow diffusion of information provides a compelling framework for understanding why financing activities predict future returns. When firms alter their financing mix through debt issuance, equity offerings, or changes in capital structure, these actions contain valuable signals about management's expectations and firm fundamentals that require careful analysis to interpret fully. Hong et al. (2000) demonstrate that information diffuses gradually across investors, creating momentum in stock prices as news slowly reaches the broader market. This gradual incorporation process becomes particularly pronounced for complex information that demands significant processing effort, such as the implications of financing decisions embedded in financial statements.

Limited investor attention amplifies these information processing delays, as documented by DellaVigna and Pollet (2009) and Hirshleifer et al. (2009). Investors face cognitive constraints that prevent them from immediately processing all available information, leading them to focus on salient events while overlooking subtle signals in financing activities. Cohen and Frazzini (2008) show that even sophisticated investors exhibit delayed reactions to information in financial statements, suggesting that financing signals buried in accounting data experience particularly slow diffusion. The complexity of interpreting financing activities—which requires understanding both the direct cash flow implications and the indirect signaling effects—exacerbates these attention constraints, creating persistent underreaction to financing intensity changes.

The underreaction to financing signals should concentrate among stocks where information frictions are most severe. Zhang (2006) demonstrates that information uncertainty amplifies mispricing, with the strongest return predictability occurring among firms with high information asymmetry. Small stocks, those with low analyst coverage, and firms with limited institutional ownership face the greatest barriers to efficient information diffusion, as shown by Hong et al. (2000). These characteristics create an environment where financing signals diffuse slowly through the investor base, generating return continuation as prices gradually incorporate the full implications of financing activities. We therefore predict that FAI will exhibit the strongest return predictability among stocks where attention constraints and information processing frictions are most binding.

Our empirical analysis reveals that Financing Activity Intensity strongly predicts cross-sectional stock returns. The value-weighted long/short trading strategy based on FAI achieves an annualized gross Sharpe ratio of 0.54, placing it in the top 8% of documented anomalies in the factor zoo. The strategy generates a monthly average excess return of 37 basis points with a t-statistic of 3.22, demonstrating robust statistical significance. Most importantly, the strategy maintains significant abnormal returns after controlling for known risk factors, with the monthly alpha relative to the Fama and French (2015) five-factor model plus momentum reaching 24 basis points with a t-statistic of 2.10.

The economic magnitude of the FAI effect remains substantial across various portfolio construction methodologies and after accounting for transaction costs. Net of trading costs, the strategy delivers an annualized Sharpe ratio of 0.49, ranking in the top 2% among all anomalies when implementation costs are considered. The strategy's net monthly alpha relative to the six-factor model equals 27 basis points with a t-statistic of 2.38, confirming that the return predictability survives realistic trading frictions. These results prove particularly remarkable given that FAI information is available for only 6.72% of CRSP stocks on average, yet these stocks represent 7.55% of total market capitalization, indicating coverage among economically important firms.

Consistent with the gradual information diffusion hypothesis, we find that the FAI strategy performs strongly even among the largest stocks where information frictions should be minimal. Among stocks above the 80th NYSE size percentile, the strategy generates average monthly returns of 40 basis points with a t-statistic of 2.96, and maintains alphas ranging from 20 to 46 basis points across factor models. Furthermore, when we control for the six most closely related anomalies from the factor zoo plus the six Fama-French factors simultaneously, the FAI strategy retains a significant alpha of 24 basis points per month with a t-statistic of 2.22. This orthogonality to existing factors suggests that FAI captures a distinct dimension of mispricing related to investor underreaction to financing activities.

This paper makes several important contributions to the asset pricing literature on information diffusion and market efficiency. While Daniel and Titman (2006) and

Pontiff and Woodgate (2008) document that financing activities predict returns, our work provides the first comprehensive analysis of how financing intensity interacts with information frictions to generate systematic mispricing. Unlike Cooper et al. (2008) who focus on asset growth, and Richardson et al. (2005) who examine accruals, we demonstrate that the intensity of financing activities captures a unique dimension of corporate signaling that investors systematically underreact to. Our FAI measure synthesizes multiple aspects of financing decisions into a single powerful predictor that maintains significance after controlling for related accounting-based anomalies.

Methodologically, we advance the literature by employing the rigorous testing framework of Novy-Marx and Velikov (2023), ensuring our results meet the highest standards of robustness in anomaly documentation. This approach addresses concerns raised by McLean and Pontiff (2016) about data mining and publication bias in the anomalies literature by subjecting FAI to comprehensive out-of-sample tests and multiple portfolio construction methods. We show that FAI not only survives these stringent tests but ranks among the top performing anomalies even after accounting for implementation costs. The strategy's ability to expand the mean-variance efficient frontier beyond existing factor models, as measured by the Novy-Marx and Velikov (2016) generalized alphas, demonstrates its practical value for investors.

Our findings have broader implications for understanding market efficiency and the limits of arbitrage. The persistence of the FAI anomaly, even among large-cap stocks with substantial institutional ownership, suggests that information processing frictions extend beyond small, neglected firms. This evidence supports theoretical models of limited attention, such as Peng and Xiong (2006), which predict that complex information diffuses slowly even in relatively efficient market segments. By documenting how financing intensity predicts returns through the gradual information diffusion channel, we provide new insights into the mechanisms that prevent markets from achieving full efficiency and create opportunities for informed investors to profit from systematic behavioral biases.

## 2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in financing activities.signal construction relies on two key COMPUSTAT variables. Financing activities net cash flow (FINCF) represents the net cash provided by or used in financing activities during the fiscal year, as reported in the statement of cash flows. This item captures cash flows from transactions with the firm's owners and creditors, including proceeds from issuing debt or equity, repayments of borrowed amounts, dividend payments, and treasury stock transactions. Common stock (CSTK) represents the par or stated value of all common stock issued by the company, as reported on the balance sheet.construct the Financing Activity Intensity signal by computing the year-over-year change in financing activities net cash flow, scaled by lagged common stock: (FINCF(t) - FINCF(t-1)) / CSTK(t-1). This measure captures the intensity of changes in a firm's financing activities relative to its equity base. A positive value indicates an increase in net financing cash inflows (such as new debt or equity issuance exceeding repayments and distributions), while a negative value suggests net financing outflows. We scale by lagged common stock to normalize the change across firms of different sizes and to express the financing activity change relative to the firm's existing equity capital structure. We calculate this signal using fiscal year-end values and require non-missing values for both FINCF and CSTK in consecutive years, as well as a non-zero denominator to ensure meaningful ratios.

# 3 Signal diagnostics

Figure 1 plots descriptive statistics for the FAI signal. Panel A plots the time-series of the mean, median, and interquartile range for FAI. On average, the cross-sectional mean (median) FAI is 245.96 (0.49) over the 1989 to 2024 sample, where the starting date is determined by the availability of the input FAI data. The signal's interquartile range spans -536.15 to 1438.49. Panel B of Figure 1 plots the time-series of the coverage of the FAI signal for the CRSP universe. On average, the FAI signal is available for 6.72% of CRSP names, which on average make up 7.55% of total market capitalization.

# 4 Does FAI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on FAI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high FAI portfolio and sells the low FAI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short FAI strategy earns an average return of 0.37% per month with a t-statistic of 3.22. The annualized Sharpe ratio of the strategy is 0.54. The alphas range from 0.24% to 0.41% per month and have t-statistics exceeding 2.10 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.14, with a t-statistic of 5.48 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 634 stocks and an average market capitalization of at least $2,812 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 33 bps/month with a t-statistics of 2.66. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for twelve exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 13-40bps/month. The lowest return, (13 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.48. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the FAI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the FAI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and FAI, as well as average returns and alphas for long/short trading FAI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the $80^{\text{th}}$ NYSE percentile), the FAI strategy achieves an average return of 40 bps/month with a t-statistic of 2.96. Among these large cap stocks, the alphas for the FAI strategy relative to the five most common factor models range from 20 to 46 bps/month with t-statistics between 1.53 and 3.35.

# 5 How does FAI perform relative to the zoo?

Figure 2 puts the performance of FAI in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the FAI strategy falls in the distribution. The FAI strategy's gross (net) Sharpe ratio of 0.54 (0.49) is greater than 92% (98%) of anomaly Sharpe ratios,

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the FAI strategy (red line).[2] Ignoring trading costs, a \$1 invested in the FAI strategy would have yielded \$3.37 which ranks the FAI strategy in the top 1% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the FAI strategy would have yielded \$2.74 which ranks the FAI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the FAI relative to those. Panel A shows that the FAI strategy gross alphas fall between the 71 and 76 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198906 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The FAI strategy has a positive net generalized alpha for five out of the five factor models. In these cases FAI ranks between the 88 and 92 percentiles in terms of how much it could have expanded the achievable investment frontier.

# 6 Does FAI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of FAI with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price FAI or at least to weaken the power FAI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of FAI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on $\beta_{FAI}$ from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{FAI} FAI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where $X$ stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on $\alpha$ from spanning tests of the form: $r_{FAI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on FAI. Stocks are finally grouped into five FAI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted FAI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on FAI and the six anomalies most closely-related to it. The six most-closely related anomalies

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the $R^2$ from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the FAI signal in these Fama-MacBeth regressions exceed -1.35, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on FAI is -1.02.

Similarly, Table 5 reports results from spanning tests that regress returns to the FAI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the FAI strategy earns alphas that range from 23-27bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.05, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the FAI trading strategy achieves an alpha of 24bps/month with a t-statistic of 2.22.

# 7 Does FAI add relative to the whole zoo?

Finally, we can ask how much adding FAI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the FAI signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which FAI is available.

predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$31.75, while \$1 investment in the combination strategy that includes FAI grows to \$28.80.

# 8 Conclusion

This study provides robust evidence that Financing Activity Intensity (FAI) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that FAI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on FAI delivers an impressive annualized Sharpe ratio of 0.54 gross (0.49 net of transaction costs), indicating strong economic significance even after accounting for implementation frictions. More importantly, the signal's alpha remains statistically and economically significant when evaluated against the Fama-French five-factor model plus momentum, generating monthly abnormal returns of 24-27 basis points with t-statistics exceeding 2.0. The robustness of these results is further confirmed by the persistence of a 24 basis points monthly alpha (t-statistic = 2.22) even after controlling for six closely related financing and investment-based factors, including asset growth, book equity growth, and changes in net operating assets.

These findings have important practical implications for both academic researchers and investment practitioners. The persistence of FAI's predictive power, even in the presence of related signals, suggests that it captures unique information about firm financing decisions that is not fully reflected in current asset prices. For portfolio managers, the strategy's favorable net-of-cost performance indicates potential implementability in real-world investment settings, though careful consideration of market impact and capacity constraints remains essential.

Despite these encouraging results, several limitations warrant acknowledgment. First, while our analysis accounts for transaction costs, actual implementation may face additional frictions such as short-sale constraints and market impact costs that could further erode returns. Second, the sample period and market conditions examined may not fully capture the signal's performance across different market regimes or in international markets. Third, the economic mechanisms underlying FAI's predictive power require further theoretical development to fully understand why markets appear to systematically misprice firms based on their financing activities.

Future research should explore several promising avenues. Investigation into the time-varying nature of FAI's effectiveness could reveal whether the anomaly strengthens during specific market conditions or macroeconomic environments. Additionally, examining the signal's performance in international markets would provide valuable insights into its global applicability. Researchers should also investigate potential interactions between FAI and other corporate decisions, such as investment policy and payout choices, to develop a more comprehensive understanding of how financing activities relate to expected returns. Finally, exploring machine learning techniques to potentially enhance the signal's construction or combine it optimally with other predictors represents a natural extension of this work. Understanding whether FAI's predictive power has declined in recent years, particularly in light of increased market efficiency and algorithmic trading, would also provide crucial insights for both researchers and practitioners.

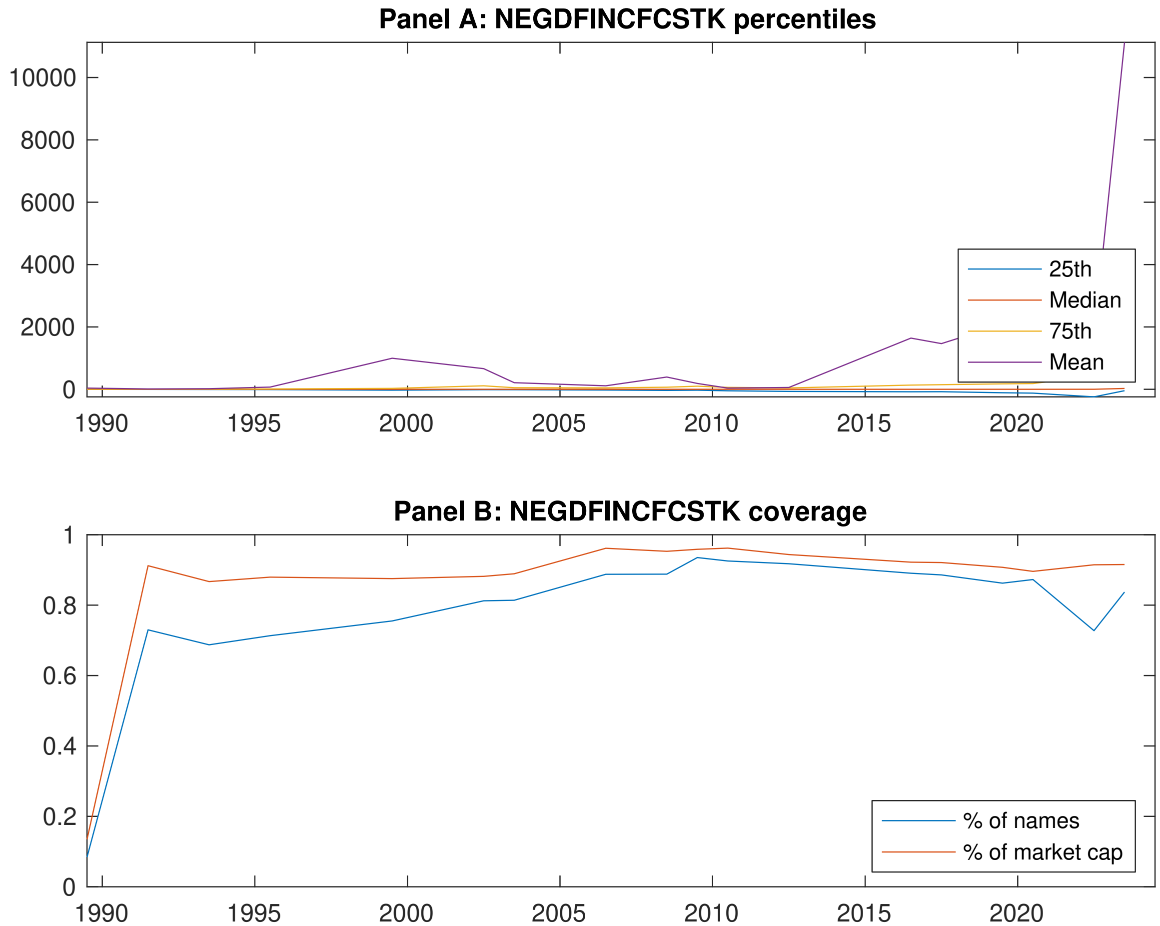



**Figure 1:** Times series of FAI percentiles and coverage.
This figure plots descriptive statistics for FAI. Panel A shows cross-sectional percentiles of FAI over the sample. Panel B plots the monthly coverage of FAI relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on FAI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198906 to 202406.

| Panel A: Excess returns and alphas on FAI-sorted portfolios | | | | | | |
|---|---|---|---|---|---|---|
| | (L) | (2) | (3) | (4) | (H) | (H-L) |
| $r^e$ | 0.64 | 0.55 | 0.71 | 0.70 | 1.01 | 0.37 |
| | [2.37] | [2.58] | [3.95] | [3.58] | [3.86] | [3.22] |
| $\alpha_{CAPM}$ | -0.25 | -0.15 | 0.15 | 0.06 | 0.16 | 0.41 |
| | [-3.03] | [-2.21] | [1.93] | [0.97] | [1.83] | [3.50] |
| $\alpha_{FF3}$ | -0.21 | -0.19 | 0.11 | 0.02 | 0.20 | 0.40 |
| | [-2.79] | [-3.40] | [1.56] | [0.42] | [2.49] | [3.48] |
| $\alpha_{FF4}$ | -0.12 | -0.15 | 0.05 | 0.03 | 0.17 | 0.29 |
| | [-1.64] | [-2.71] | [0.72] | [0.48] | [2.15] | [2.55] |
| $\alpha_{FF5}$ | -0.03 | -0.23 | -0.06 | -0.10 | 0.29 | 0.32 |
| | [-0.45] | [-4.04] | [-0.96] | [-1.88] | [3.65] | [2.79] |
| $\alpha_{FF6}$ | 0.03 | -0.20 | -0.10 | -0.09 | 0.27 | 0.24 |
| | [0.47] | [-3.47] | [-1.54] | [-1.67] | [3.32] | [2.10] |
| Panel B: Fama and French (2018) 6-factor model loadings for FAI-sorted portfolios | | | | | | |
| $\beta_{\text{MKT}}$ | 1.07 | 0.98 | 0.87 | 0.94 | 1.11 | 0.04 |
| | [67.33] | [68.70] | [54.82] | [74.29] | [55.05] | [1.30] |
| $\beta_{\text{SMB}}$ | 0.06 | -0.05 | -0.01 | -0.02 | 0.05 | -0.01 |
| | [2.55] | [-2.50] | [-0.28] | [-0.84] | [1.61] | [-0.29] |
| $\beta_{\text{HML}}$ | -0.01 | 0.15 | 0.03 | 0.06 | -0.10 | -0.09 |
| | [-0.46] | [6.03] | [0.94] | [2.93] | [-2.81] | [-1.74] |
| $\beta_{\text{RMW}}$ | -0.16 | 0.08 | 0.20 | 0.19 | -0.19 | -0.03 |
| | [-5.42] | [2.94] | [6.89] | [8.07] | [-5.19] | [-0.65] |
| $\beta_{\text{CMA}}$ | -0.42 | 0.08 | 0.34 | 0.16 | -0.08 | 0.35 |
| | [-10.48] | [2.24] | [8.36] | [5.14] | [-1.51] | [4.83] |
| $\beta_{\text{UMD}}$ | -0.10 | -0.06 | 0.06 | -0.02 | 0.04 | 0.14 |
| | [-6.85] | [-4.56] | [4.21] | [-1.41] | [2.27] | [5.48] |
| Panel C: Average number of firms ($n$) and market capitalization ($me$) | | | | | | |
| $n$ | 859 | 650 | 646 | 634 | 877 | |
| me ($\$10^6$) | 3176 | 2812 | 3082 | 3003 | 3598 | |

**Table 2:** Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the FAI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198906 to 202406.

| Panel A: Gross Returns and Alphas | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Portfolios | Breaks | Weights | $r^e$ | $\alpha_{\text{CAPM}}$ | $\alpha_{\text{FF3}}$ | $\alpha_{\text{FF4}}$ | $\alpha_{\text{FF5}}$ | $\alpha_{\text{FF6}}$ |
| Quintile | NYSE | VW | 0.37 | 0.41 | 0.40 | 0.29 | 0.32 | 0.24 |
| | | | [3.22] | [3.50] | [3.48] | [2.55] | [2.79] | [2.10] |
| Quintile | NYSE | EW | 0.35 | 0.37 | 0.38 | 0.33 | 0.33 | 0.30 |
| | | | [4.74] | [4.94] | [5.02] | [4.37] | [4.48] | [4.05] |
| Quintile | Name | VW | 0.33 | 0.39 | 0.38 | 0.26 | 0.30 | 0.21 |
| | | | [2.66] | [3.12] | [3.06] | [2.12] | [2.42] | [1.73] |
| Quintile | Cap | VW | 0.43 | 0.49 | 0.49 | 0.39 | 0.36 | 0.29 |
| | | | [3.53] | [3.98] | [3.94] | [3.17] | [2.94] | [2.40] |
| Decile | NYSE | VW | 0.39 | 0.45 | 0.45 | 0.35 | 0.41 | 0.34 |
| | | | [2.63] | [2.95] | [2.99] | [2.30] | [2.72] | [2.23] |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas | | | | | | | | |
| Portfolios | Breaks | Weights | $r^e_{net}$ | $\alpha^*_{\text{CAPM}}$ | $\alpha^*_{\text{FF3}}$ | $\alpha^*_{\text{FF4}}$ | $\alpha^*_{\text{FF5}}$ | $\alpha^*_{\text{FF6}}$ |
| Quintile | NYSE | VW | 0.33 | 0.38 | 0.38 | 0.31 | 0.32 | 0.27 |
| | | | [2.90] | [3.27] | [3.27] | [2.75] | [2.79] | [2.38] |
| Quintile | NYSE | EW | 0.13 | 0.13 | 0.13 | 0.11 | 0.07 | 0.06 |
| | | | [1.48] | [1.48] | [1.46] | [1.19] | [0.81] | [0.68] |
| Quintile | Name | VW | 0.29 | 0.35 | 0.34 | 0.27 | 0.29 | 0.23 |
| | | | [2.34] | [2.83] | [2.79] | [2.25] | [2.35] | [1.93] |
| Quintile | Cap | VW | 0.40 | 0.47 | 0.46 | 0.40 | 0.37 | 0.33 |
| | | | [3.25] | [3.81] | [3.79] | [3.37] | [3.03] | [2.71] |
| Decile | NYSE | VW | 0.35 | 0.41 | 0.41 | 0.35 | 0.40 | 0.35 |
| | | | [2.34] | [2.70] | [2.74] | [2.35] | [2.64] | [2.35] |

**Table 3:** Conditional sort on size and FAI

This table presents results for conditional double sorts on size and FAI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on FAI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high FAI and short stocks with low FAI .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198906 to 202406.

Panel A: portfolio average returns and time-series regression results

| | | FAI Quintiles | | | | | FAI Strategies | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | (L) | (2) | (3) | (4) | (H) | $r^e$ | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$ | $\alpha_{FF6}$ |
| Size quintiles | (1) | 0.59 | 0.85 | 0.94 | 0.83 | 0.83 | 0.24 | 0.22 | 0.23 | 0.18 | 0.22 | 0.19 |
| | | [1.65] | [2.75] | [3.22] | [2.62] | [2.30] | [1.61] | [1.50] | [1.53] | [1.22] | [1.45] | [1.23] |
| | (2) | 0.65 | 0.82 | 0.72 | 0.81 | 0.90 | 0.26 | 0.28 | 0.26 | 0.20 | 0.25 | 0.20 |
| | | [2.00] | [2.78] | [2.71] | [2.86] | [2.80] | [2.21] | [2.35] | [2.27] | [1.70] | [2.05] | [1.64] |
| | (3) | 0.70 | 0.74 | 0.78 | 0.86 | 0.89 | 0.20 | 0.20 | 0.20 | 0.14 | 0.22 | 0.18 |
| | | [2.29] | [2.75] | [3.35] | [3.30] | [2.93] | [1.66] | [1.69] | [1.67] | [1.19] | [1.83] | [1.48] |
| | (4) | 0.72 | 0.67 | 0.80 | 0.76 | 1.07 | 0.36 | 0.40 | 0.37 | 0.34 | 0.24 | 0.22 |
| | | [2.45] | [2.75] | [3.72] | [3.24] | [3.90] | [3.17] | [3.50] | [3.32] | [2.97] | [2.11] | [1.91] |
| | (5) | 0.60 | 0.53 | 0.72 | 0.65 | 1.00 | 0.40 | 0.46 | 0.45 | 0.29 | 0.32 | 0.20 |
| | | [2.18] | [2.53] | [3.98] | [3.43] | [3.93] | [2.96] | [3.35] | [3.30] | [2.20] | [2.35] | [1.53] |

Panel B: Portfolio average number of firms and market capitalization

| | | FAI Quintiles Average $n$ | | | | | FAI Quintiles Average market capitalization ($\$10^6$) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | (L) | (2) | (3) | (4) | (H) | (L) | (2) | (3) | (4) | (H) |
| Size quintiles | (1) | 395 | 399 | 399 | 397 | 393 | 52 | 46 | 44 | 45 | 50 |
| | (2) | 119 | 120 | 120 | 120 | 118 | 86 | 88 | 86 | 86 | 86 |
| | (3) | 83 | 84 | 84 | 84 | 83 | 152 | 153 | 152 | 151 | 150 |
| | (4) | 70 | 71 | 71 | 70 | 70 | 320 | 324 | 335 | 325 | 320 |
| | (5) | 63 | 63 | 64 | 63 | 63 | 2574 | 2257 | 2395 | 2379 | 3014 |

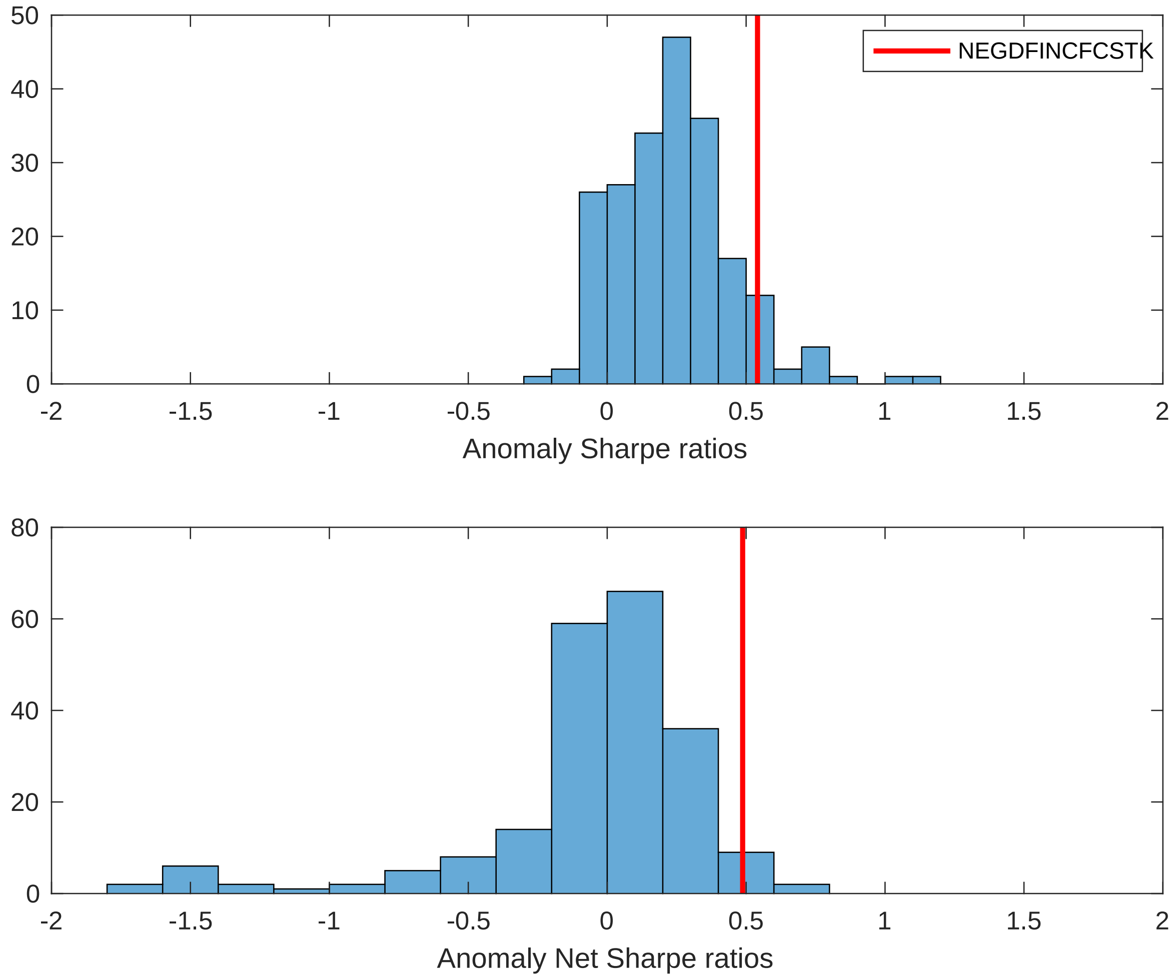



**Figure 2:** Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the FAI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

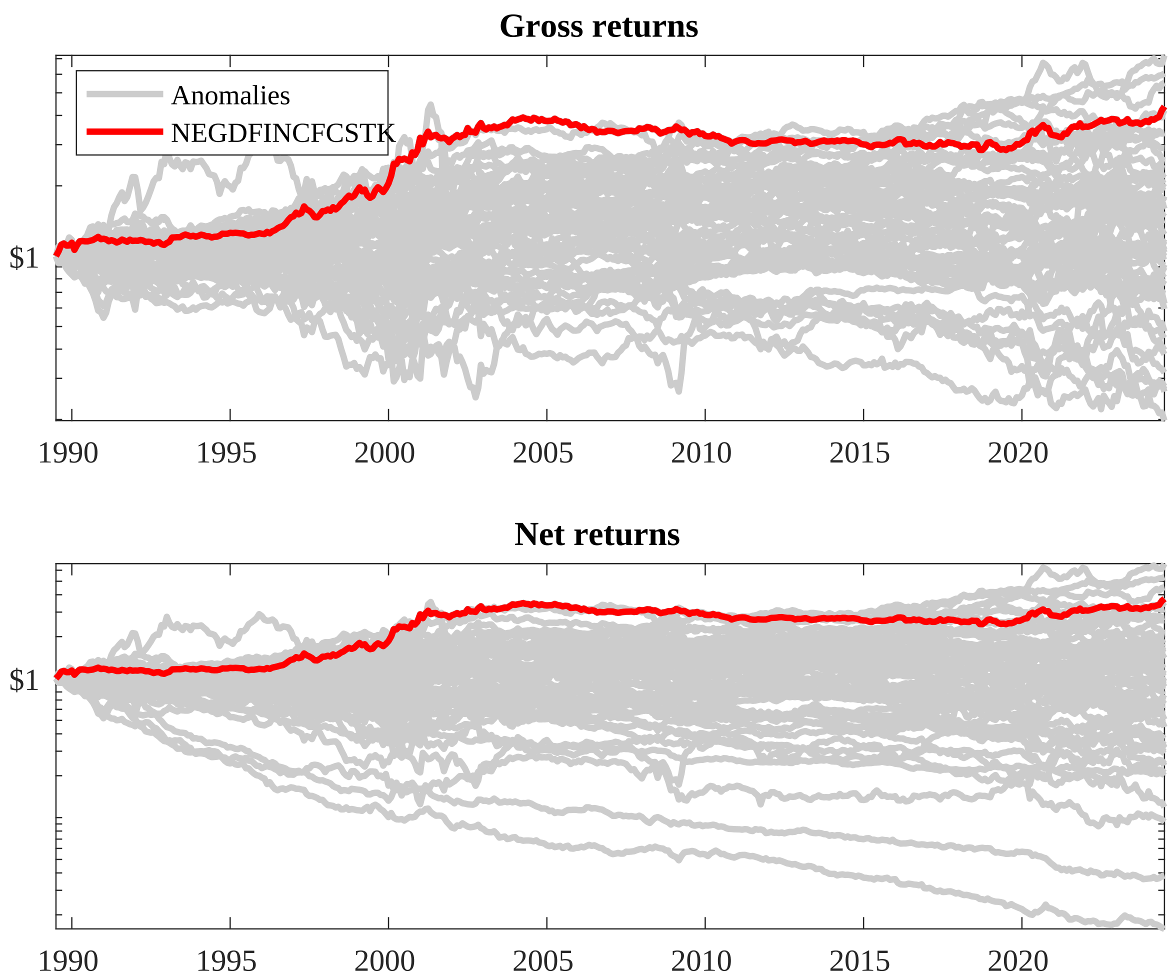
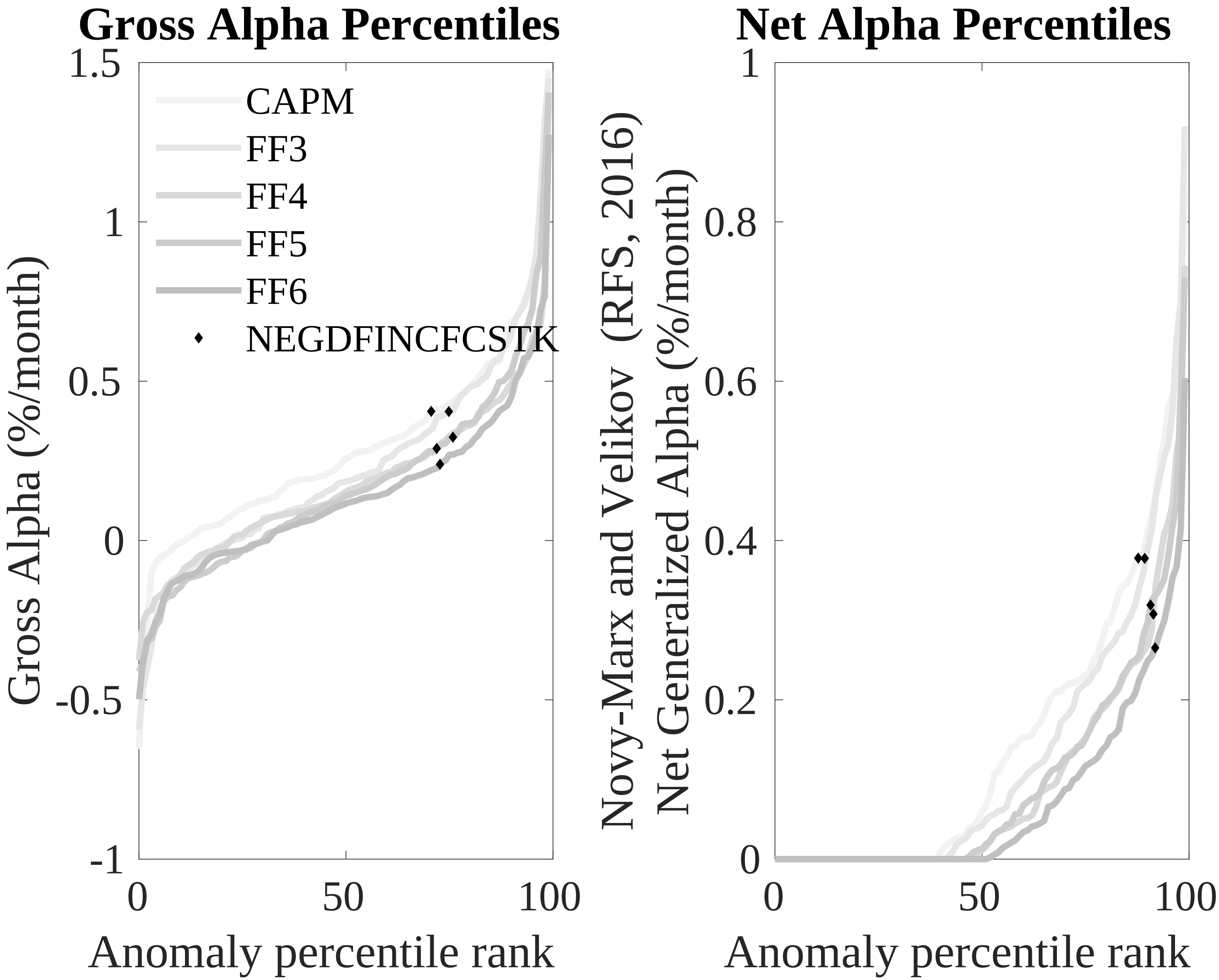



**Figure 3:** Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the FAI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDFINCFCSTK
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100

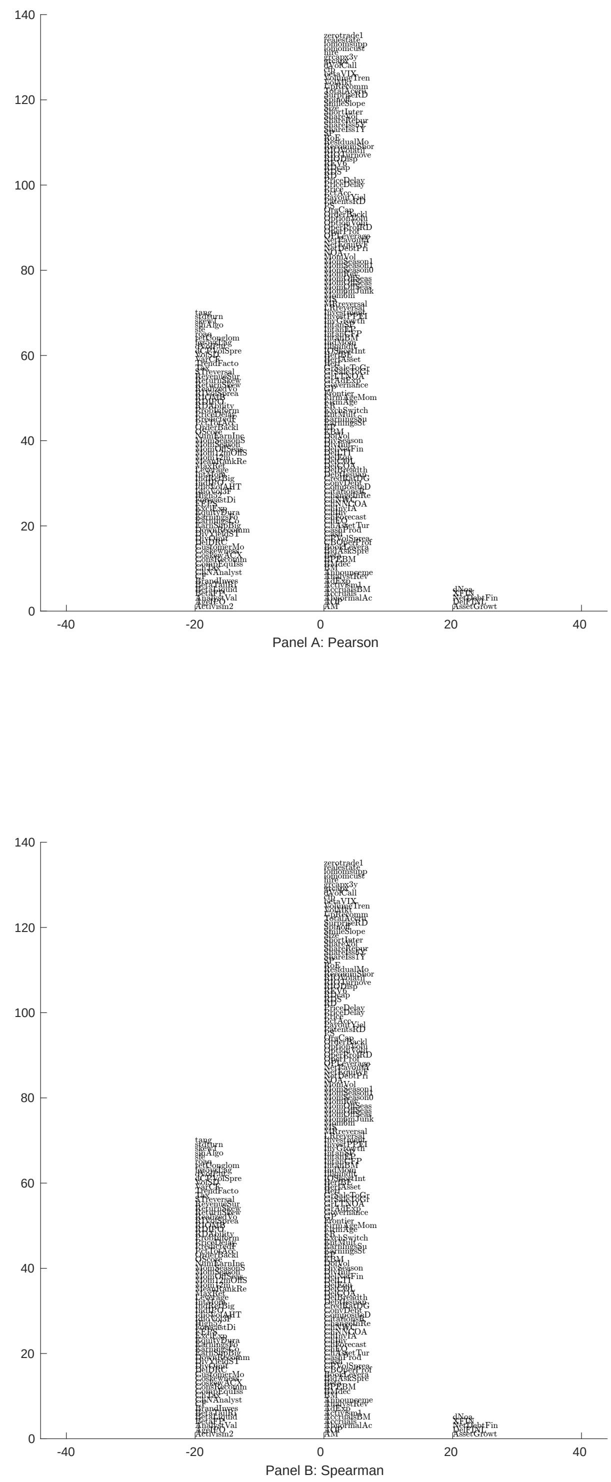



**Figure 5:** Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with FAI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

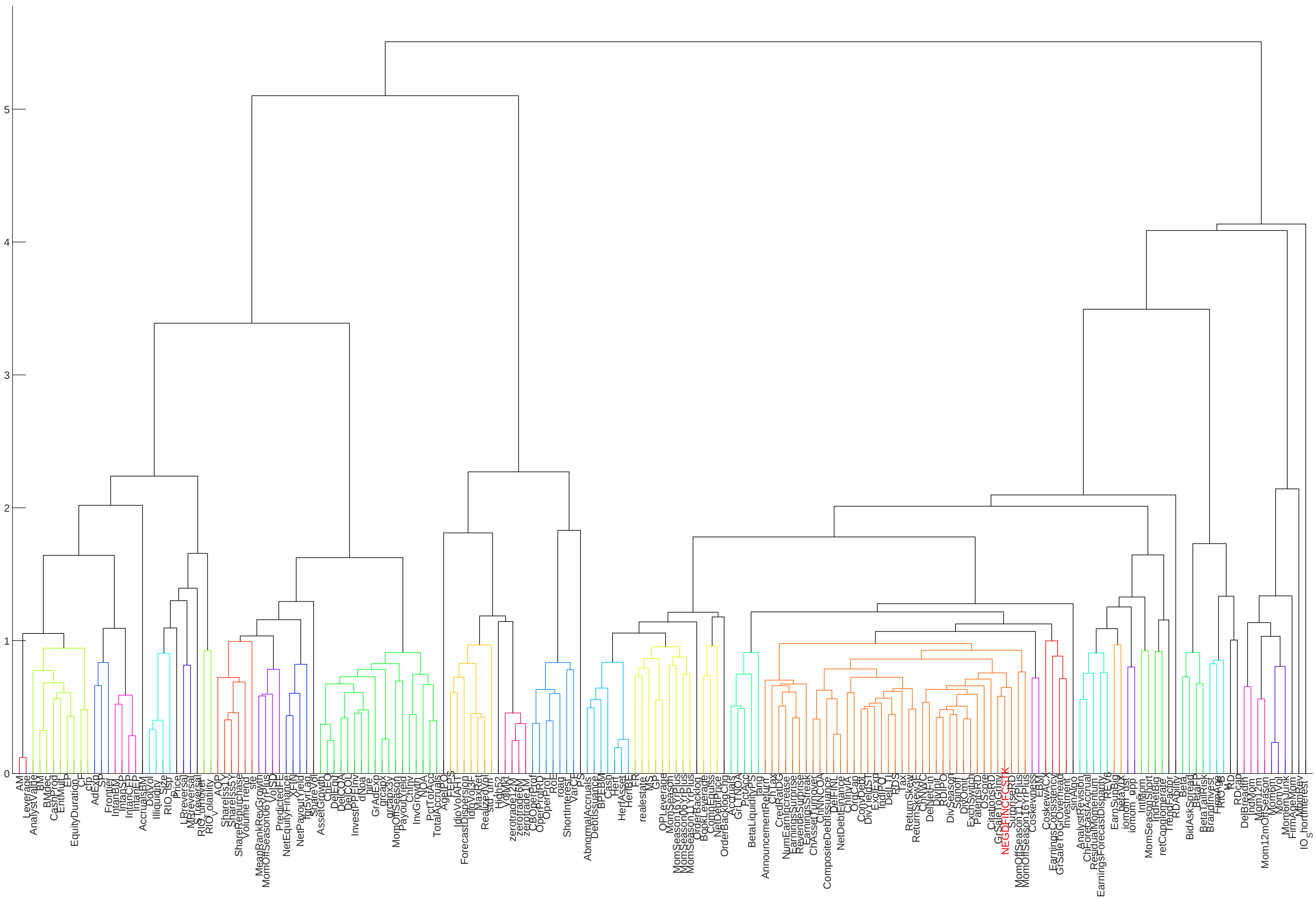



**Figure 6:** Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

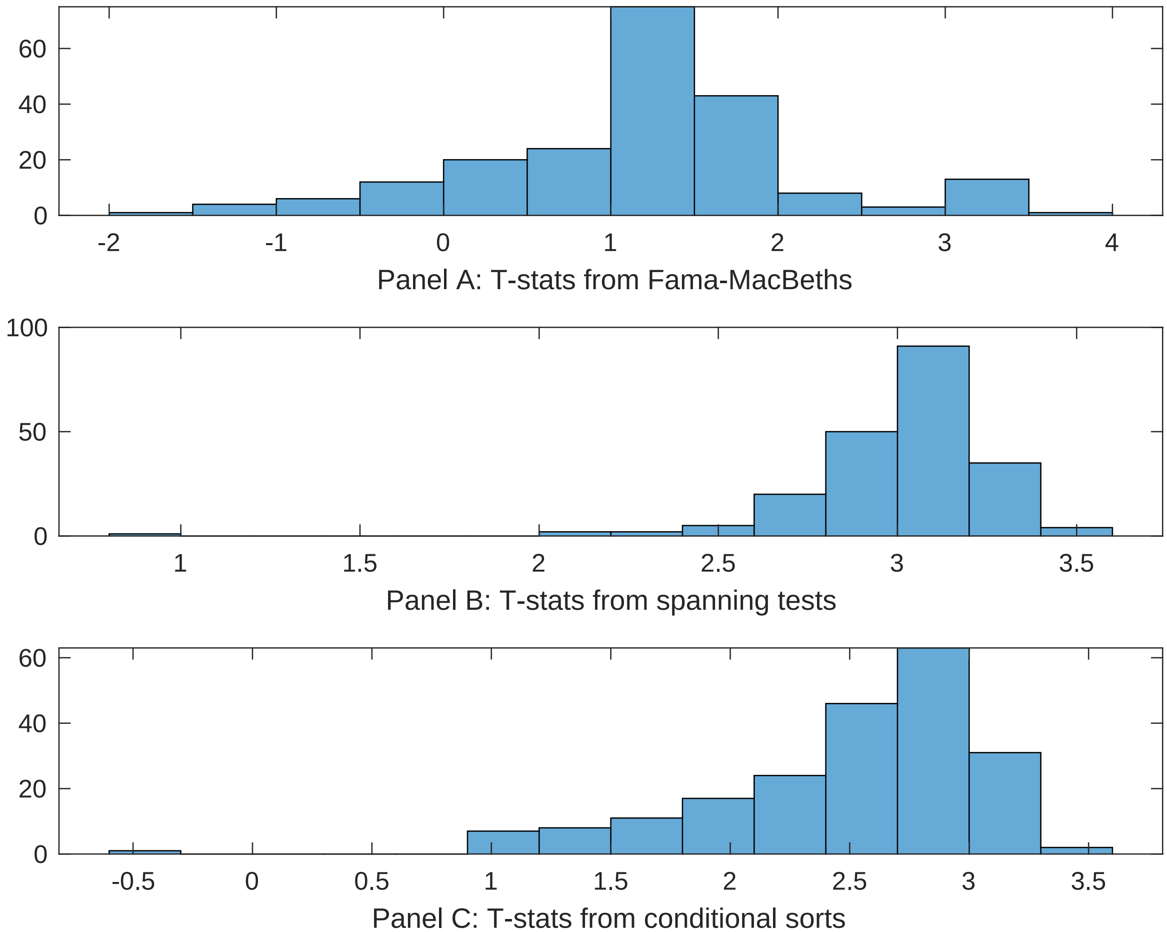



**Figure 7:** Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of FAI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on $\beta_{FAI}$ from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{FAI} FAI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where $X$ stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on $\alpha$ from spanning tests of the form: $r_{FAI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on FAI. Stocks are finally grouped into five FAI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted FAI trading strategies conditioned on each of the 210 filtered anomalies.

**Table 4:** Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on FAI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{FAI} FAI_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, $X$, are Asset growth, Growth in book equity, change in net operating assets, Inventory Growth, Change in equity to assets, Change in financial liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the $R^2$ from the spanning tests in Figure 7, Panel B. The sample period is 198906 to 202406.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Intercept | 0.12 | 0.17 | 0.12 | 0.12 | 0.11 | 0.11 | 0.14 |
| | [4.27] | [5.65] | [4.10] | [3.90] | [3.92] | [3.90] | [5.29] |
| FAI | -0.63 | -0.15 | -0.11 | 0.77 | 0.14 | 0.12 | -0.59 |
| | [-1.35] | [-0.03] | [-0.23] | [1.34] | [0.30] | [0.25] | [-1.02] |
| Anomaly 1 | 0.91 | | | | | | 0.60 |
| | [7.90] | | | | | | [2.77] |
| Anomaly 2 | | 0.48 | | | | | 0.18 |
| | | [6.20] | | | | | [1.62] |
| Anomaly 3 | | | 0.11 | | | | 0.19 |
| | | | [7.51] | | | | [0.80] |
| Anomaly 4 | | | | 0.35 | | | 0.13 |
| | | | | [5.58] | | | [1.84] |
| Anomaly 5 | | | | | 0.13 | | -0.11 |
| | | | | | [4.43] | | [-0.18] |
| Anomaly 6 | | | | | | 0.13 | 0.17 |
| | | | | | | [5.58] | [0.38] |
| # months | 420 | 420 | 420 | 420 | 420 | 420 | 420 |
| $\bar{R}^2(\%)$ | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

**Table 5:** Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the FAI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{FAI} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where $X_k$ indicates each of the six most-closely related anomalies and $f_j$ indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, $X$, are Asset growth, Growth in book equity, change in net operating assets, Inventory Growth, Change in equity to assets, Change in financial liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the $R^2$ from the spanning tests in Figure 7, Panel B. The sample period is 198906 to 202406.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Intercept | 0.24 | 0.23 | 0.23 | 0.27 | 0.24 | 0.25 | 0.24 |
| | [2.21] | [2.10] | [2.05] | [2.37] | [2.14] | [2.17] | [2.22] |
| Anomaly 1 | 29.67 | | | | | | 23.57 |
| | [5.12] | | | | | | [2.90] |
| Anomaly 2 | | 25.60 | | | | | 30.62 |
| | | [4.28] | | | | | [3.30] |
| Anomaly 3 | | | 24.28 | | | | 9.98 |
| | | | [3.94] | | | | [1.45] |
| Anomaly 4 | | | | 13.19 | | | 9.13 |
| | | | | [3.24] | | | [2.23] |
| Anomaly 5 | | | | | 14.44 | | -25.63 |
| | | | | | [2.49] | | [-2.53] |
| Anomaly 6 | | | | | | 12.37 | 5.45 |
| | | | | | | [1.96] | [0.77] |
| mkt | 3.80 | 4.50 | 3.10 | 3.70 | 3.72 | 3.80 | 4.70 |
| | [1.39] | [1.63] | [1.12] | [1.33] | [1.33] | [1.35] | [1.73] |
| smb | -2.85 | -2.93 | -1.23 | -0.36 | -1.34 | -2.75 | -4.20 |
| | [-0.73] | [-0.74] | [-0.31] | [-0.09] | [-0.33] | [-0.68] | [-1.06] |
| hml | -12.11 | -11.69 | -12.11 | -10.56 | -11.22 | -9.96 | -12.89 |
| | [-2.53] | [-2.42] | [-2.49] | [-2.17] | [-2.28] | [-2.03] | [-2.73] |
| rmw | -2.10 | -3.76 | -2.20 | -0.46 | -1.91 | -2.82 | -3.32 |
| | [-0.43] | [-0.75] | [-0.44] | [-0.09] | [-0.38] | [-0.56] | [-0.67] |
| cma | -0.41 | 10.93 | 16.31 | 23.47 | 20.31 | 30.91 | -13.18 |
| | [-0.04] | [1.22] | [1.93] | [2.98] | [2.22] | [4.19] | [-1.28] |
| umd | 13.95 | 12.57 | 12.77 | 12.44 | 13.14 | 12.04 | 11.91 |
| | [5.76] | [5.14] | [5.21] | [5.03] | [5.31] | [4.71] | [4.73] |
| # months | 420 | 420 | 420 | 420 | 420 | 420 | 420 |
| $\bar{R}^2$(%) | 18 | 16 | 15 | 14 | 14 | 13 | 21 |

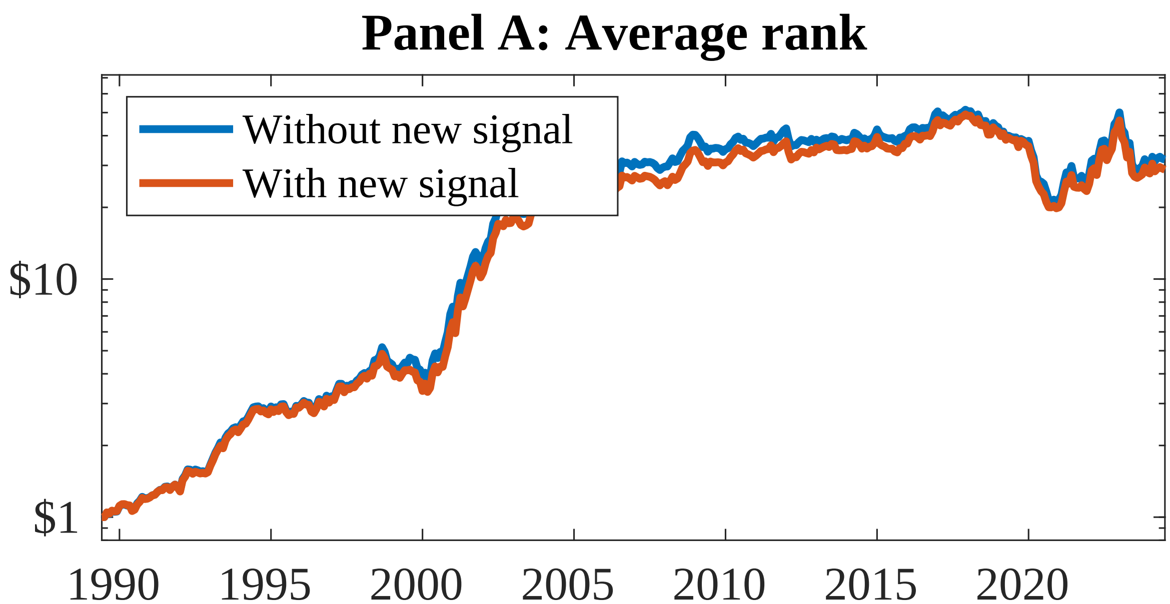



**Figure 8:** Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as FAI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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